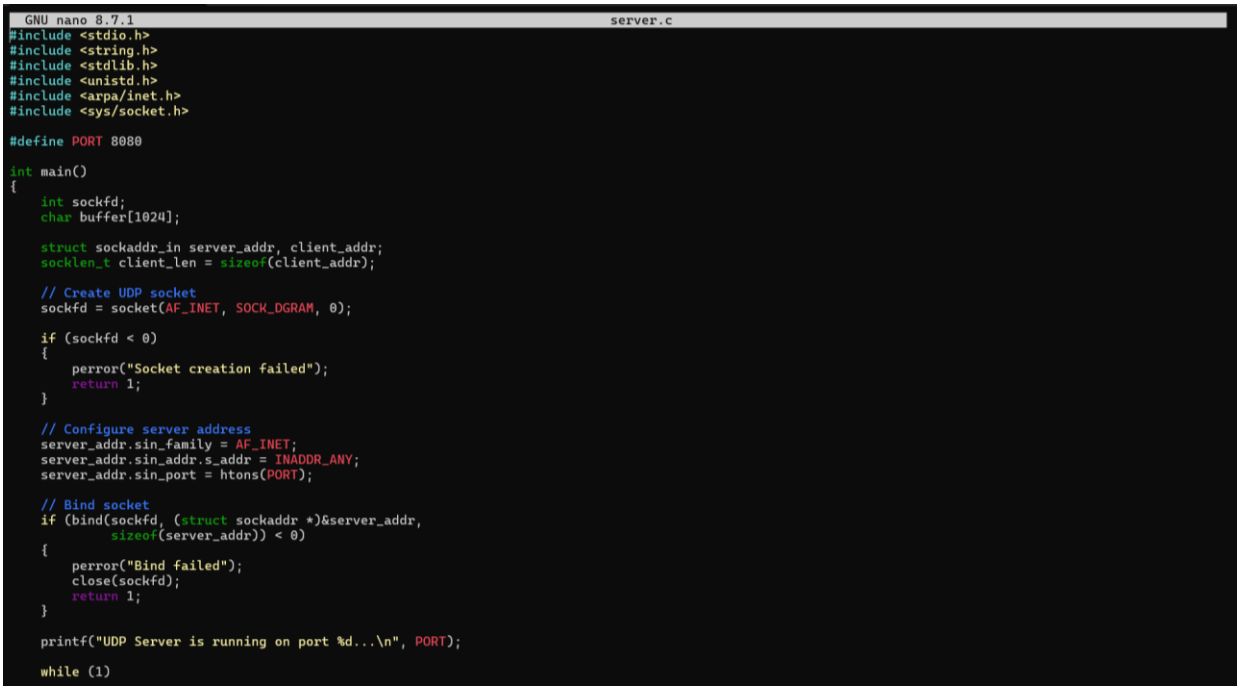
Name: Shreya Bawankar

Prn: 24070521120

Sec: B Batch: B2



```
 GNU nano 8.7.1                                server.c
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <sys/socket.h>

#define PORT 8080

int main()
{
    int sockfd;
    char buffer[1024];

    struct sockaddr_in server_addr, client_addr;
    socklen_t client_len = sizeof(client_addr);

    // Create UDP socket
    sockfd = socket(AF_INET, SOCK_DGRAM, 0);

    if (sockfd < 0)
    {
        perror("Socket creation failed");
        return 1;
    }

    // Configure server address
    server_addr.sin_family = AF_INET;
    server_addr.sin_addr.s_addr = INADDR_ANY;
    server_addr.sin_port = htons(PORT);

    // Bind socket
    if (bind(sockfd, (struct sockaddr *)&server_addr,
             sizeof(server_addr)) < 0)
    {
        perror("Bind failed");
        close(sockfd);
        return 1;
    }

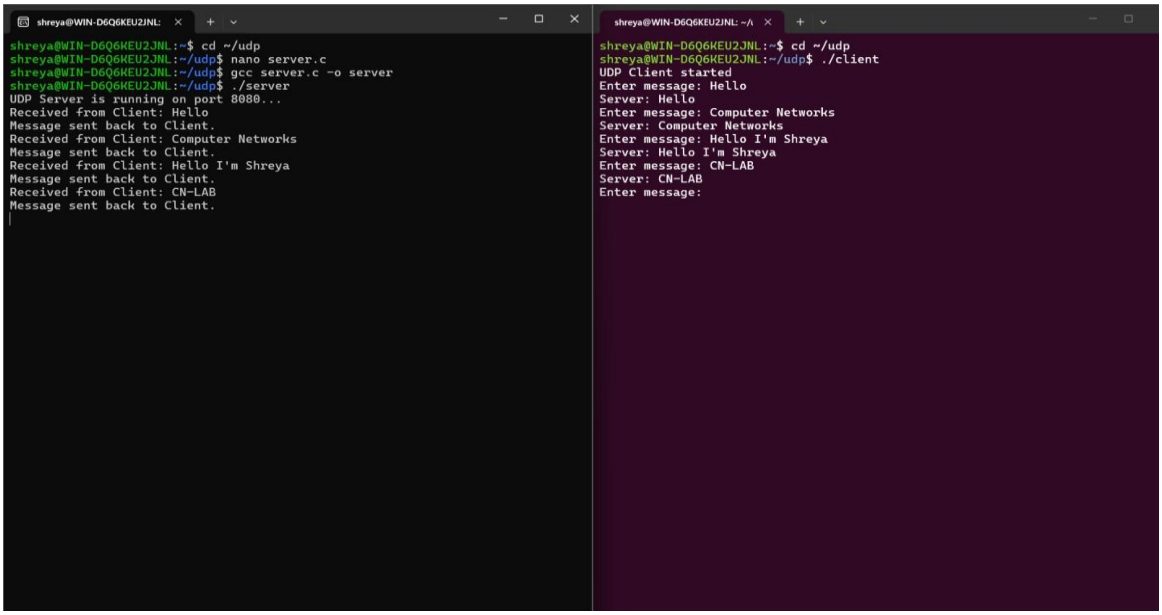
    printf("UDP Server is running on port %d...\n", PORT);

    while (1)
```

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    if (bind(sockfd, (struct sockaddr *)&server_addr,
             sizeof(server_addr)) < 0)
    {
        perror("Bind failed");
        close(sockfd);
        return 1;
    }

    printf("UDP Server is running on port %d...\n", PORT);

    while (1)
    {
        memset(buffer, 0, sizeof(buffer));

        // Receive message from client
        recvfrom(sockfd, buffer, sizeof(buffer) - 1, 0,
                 (struct sockaddr *)&client_addr, &client_len);

        printf("Received from Client: %s\n", buffer);

        // Send the same message back to client
        sendto(sockfd, buffer, strlen(buffer), 0,
               (struct sockaddr *)&client_addr, client_len);

        printf("Message sent back to Client.\n");
    }

    close(sockfd);

    return 0;
}
```